

has become significant over the last decade," says Raghuvver.

In the coming years, though, India will become a key market for offshoring end-to-end development as more focus is put on India-specific innovations and every OEM wants to make a cost-effective product for emerging markets such as India and China.

"In the past few years, many chip companies have gone on an aggressive plan to expand their presence in India to not only tap the vast talent pool but also cater to the growing demand of local consumers. In fact, as per reports, India's electronics consumption was worth \$45 billion in 2009 and is expected to increase to \$125 billion in 2014 and \$400 billion by 2020. Many chip design companies are today providing complete end-to-end solutions for not only the global market but also actively working with original equipment manufacturing partners to develop products specifically targeting the large share of local market," shares Guruswamy Ganesh, vice president and country manager, Freescale Semiconductor India.

"Growth in the hardware development is mainly happening in automotive body electronics applications and electric vehicles. In automotive development, the hardware R&D facility is generally preferred at a location near the vehicle manufacturing plant for various reasons such as testing and validation. Since hardware R&D is a continuous process, it is feasible to keep it near the manufacturing location," says Suraj Mukundarajan, director, Automotive Software Development, Infineon Technologies India.

Promising areas

Energy, defense, automotive and biomedical are the leading industries that are building embedded systems today. Over the next three years, green energy, biomedical and life sciences, and cleaner and faster transportation are poised to grow at a significantly faster pace, driving embedded system development in India.

"Automotive will be one of the

How we can do better in hardware design

For India to grab a bigger share of the hardware design market, these issues need to be addressed fast:

1. Non-availability of specialised talent
2. High initial investments. Government subsidies could come in handy
3. Organisations with huge cash reserves should invest in hardware product development ideas
4. Develop an ecosystem for manufacturing of PCBs, electronic assembly, mechanical assembly and semiconductor fabs

markets where India can be in the driver's seat. There is a huge demand locally and most of the worldwide manufacturers and suppliers have already set up their R&D, manufacturing and quality assurance teams to support the ecosystem in India," feels Galvin Lian, senior manager, sales, Fujitsu Semiconductor Asia.

India will need to largely focus on indigenisation in the defence space. It cannot afford to have embedded systems' import bills, considering the growing requirements which may even exceed fuel imports bill. We have already seen some outstanding indigenous work in defense, such as aircraft 'Tejas' of Indian Air Force and Agni-5 missile, where embedded system has played a crucial role.

India has some unique challenges due to its density and diversity of population, economy and unique market dynamics. The embedded systems built for India to address the basic needs such as energy, health and transportation have to be accessible and affordable to the common people to make them successful. These products should be reliable and robust for use in Indian environmental conditions.

We are already seeing customisation of consumer devices to suit India-specific needs.

"Unlike defence, where India does end-to-end development, in consumer devices such as mobiles we take hardware from elsewhere and do a little bit

of software work here. The same trend is happening in tablets area. There are system integration houses which are getting tablets from our neighbour countries and completely changing the user interface to suit certain applications and content that is user-specific. This is another use-case which is not end-to-end but just software part of it," informs Venkatesh Kumaran, country manager, Wind River.

In the past few years, the focus has shifted a lot towards energy-efficient electronic devices. Semiconductor companies have been quick to encash this opportunity.

"Semiconductor companies are investing a lot in designing low-power, high-performance MCUs. The sleep current for some MCUs is of the order of nanoamperes, which is very useful in battery-operated applications like smart energy meters, UPS, LED lighting and display products, point-of-sale machines, handheld ticketing machines and automotive electronics," shares Ashwin Athani, director, NGX Technologies.

Technological trends shaping up favourably

"Designing hardware is not as capital-intensive as it used to be, since advanced tools that aid in design are easily available. There is no need to design and validate expensive ASICs anymore. Most of the low-power embedded design can be done on general-purpose processors. High-end embedded systems would be using programmable/network processors or multi-core processors. Silicon vendors are providing highly integrated SoC solutions, thereby making the development of systems much more efficient," says Raghuvver.

"The vast majority of the embedded processor market remains MCUs, which are low-cost autonomous embedded processing units used in embedded applications that have lower-end processing requirements. Embedded CPUs, DSPs and FPGAs all typically provide higher levels of processing performance than MCUs